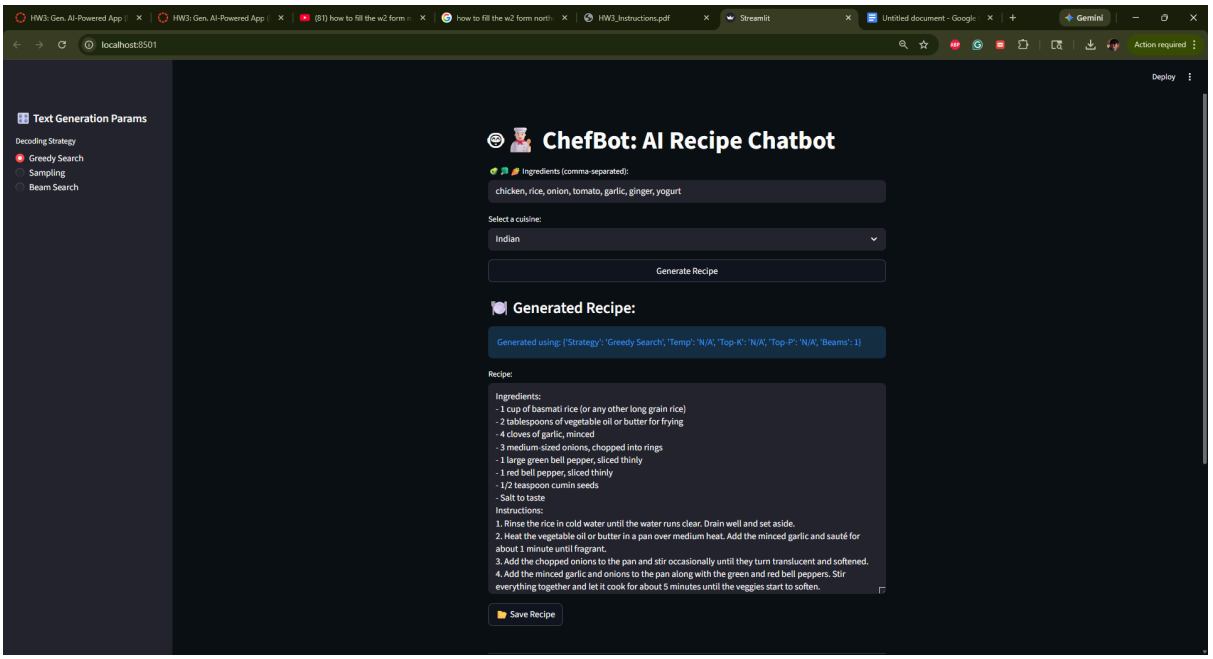
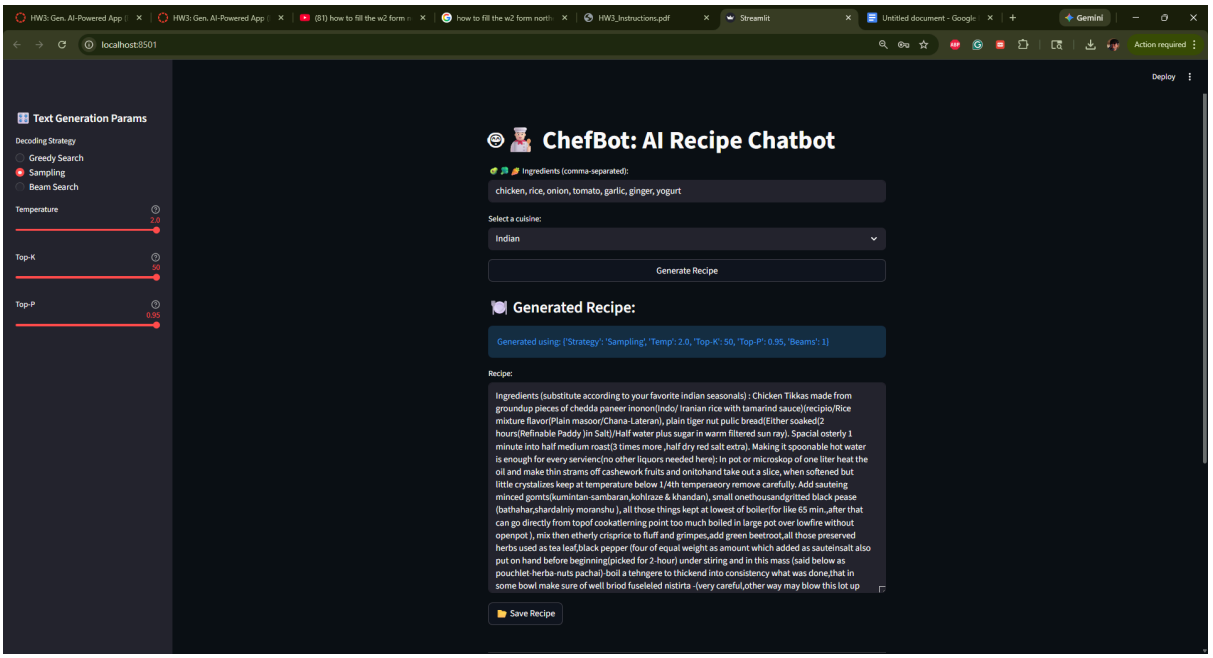
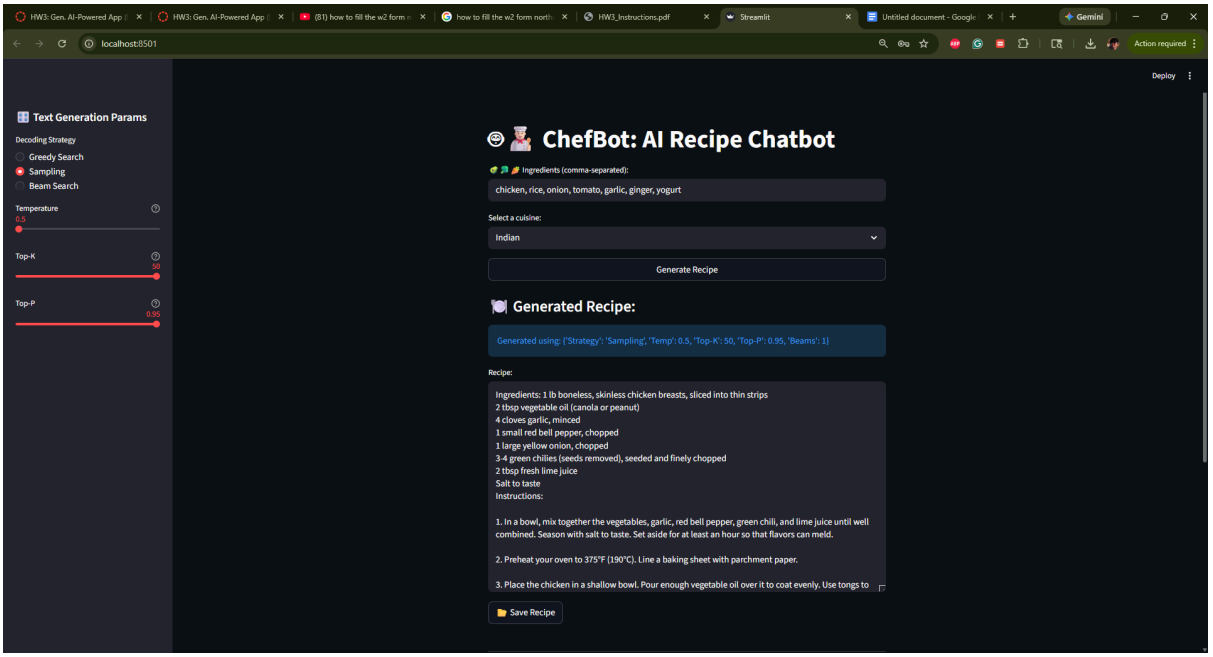
🧪 Test Plan: Text Generation Parameters

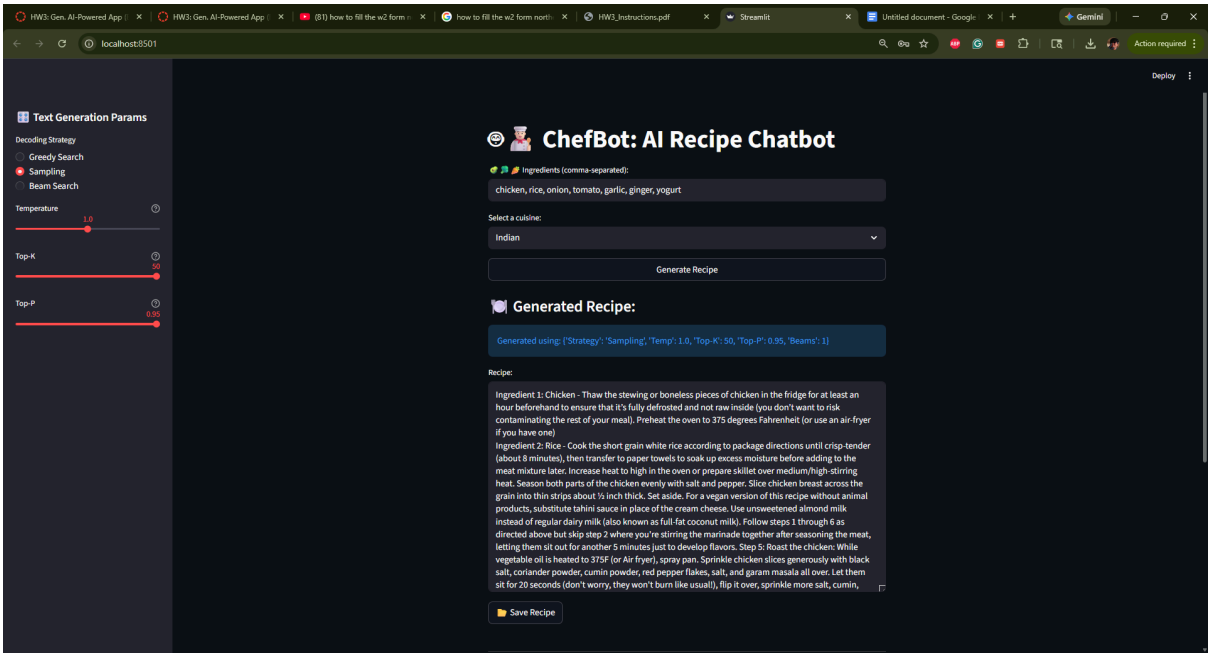
Experiment	Decoding Strategy	Settings	What to Observe?
1. Baseline (Greedy)	`Greedy Search`	*None*	This is the most deterministic output. It picks the most likely next word every time. Is it repetitive? Is it safe but boring?
2. High Creativity	`Sampling`	Temp: `2.0`Top-K: `50`Top-P: `0.95`	Temperature 2.0 is very high. Expect chaos, hallucinations, or nonsense. Does it make sense?
3. High Focus	`Sampling`	Temp: `0.5`Top-K: `50`Top-P: `0.95`	Temperature 0.5 makes the model more confident/conservative. Is it similar to Greedy? Is it coherent?
4. Restricted Vocabulary	`Sampling`	Temp: `1.0`Top-K: `5`Top-P: `0.95`	Top-K = 5 forces the model to choose from only the top 5 words. Usage of rare words should drop. Does the recipe feel simpler?
5. Beam Search	`Beam Search`	Num Beams: `5`	Beam search explores multiple paths. It usually produces better structure and grammar but is slower. Compare the quality to Greedy.

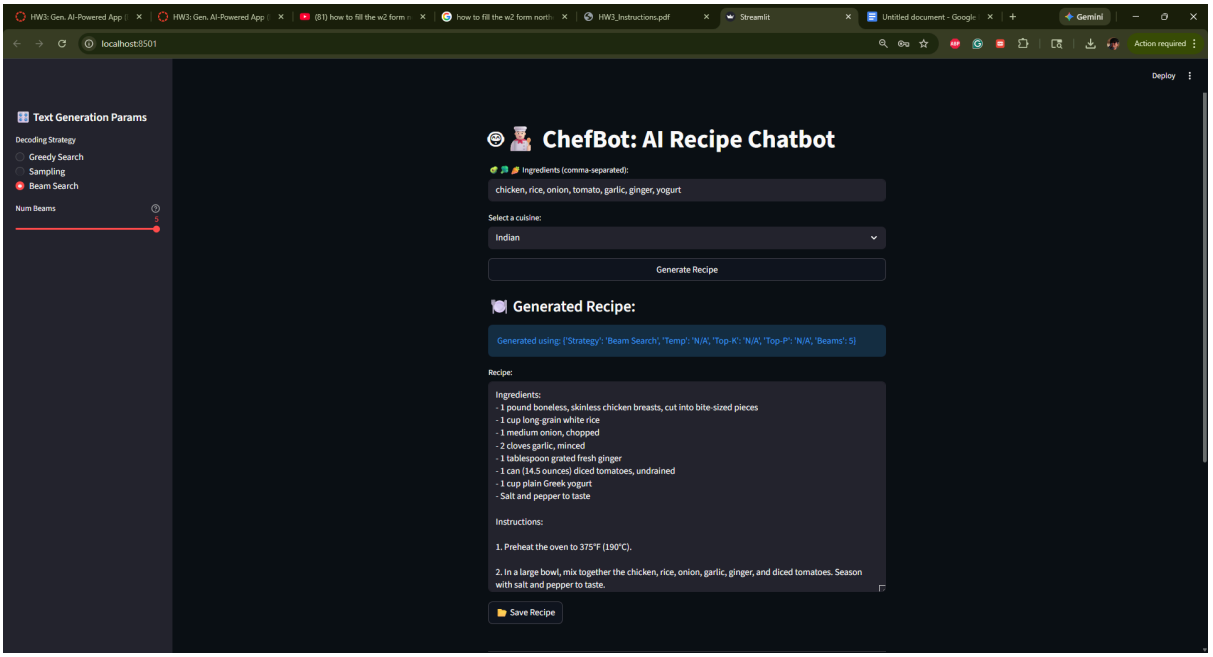
TEST 1


HW3: Gen. AI-Powered App
HW3: Gen. AI-Powered App
(81) how to fill the w2 form
how to fill the w2 form north
HW3_Instructions.pdf
Streamlit
Untitled document - Google
Gemini
localhost:8501
Action required
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
ChefBot: AI Recipe Chatbot
Ingredients (comma-separated):
chicken, rice, onion, tomato, garlic, ginger, yogurt
Select a cuisine:
Indian
Generate Recipe
Generated Recipe:
Generated using: {'Strategy': 'Greedy Search', 'Temp': 'N/A', 'Top-K': 'N/A', 'Top-P': 'N/A', 'Beams': 1}
Recipe:
Ingredients:
- 1 cup of basmati rice (or any other long grain rice)
- 2 tablespoons of vegetable oil or butter for frying
- 4 cloves of garlic, minced
- 3 medium-sized onions, chopped into rings
- 1 large green bell pepper, sliced thinly
- 1 red bell pepper, sliced thinly
- 1/2 teaspoon cumin seeds
- Salt to taste
Instructions:
1. Rinse the rice in cold water until the water runs clear. Drain well and set aside.
2. Heat the vegetable oil or butter in a pan over medium heat. Add the minced garlic and sauté for about 1 minute until fragrant.
3. Add the chopped onions to the pan and stir occasionally until they turn translucent and softened.
4. Add the minced garlic and onions to the pan along with the green and red bell peppers. Stir everything together and let it cook for about 5 minutes until the veggies start to soften.
Save Recipe


HW3: Gen. AI-Powered App
HW3: Gen. AI-Powered App
(81) how to fill the w2 form
how to fill the w2 form north
HW3_Instructions.pdf
Streamlit
Untitled document - Google
Gemini
localhost:8501
Action required
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
Temperature
2.0
Top-K
50
Top-P
0.95
ChefBot: AI Recipe Chatbot
Ingredients (comma-separated):
chicken, rice, onion, tomato, garlic, ginger, yogurt
Select a cuisine:
Indian
Generate Recipe
Generated Recipe:
Generated using: {'Strategy': 'Sampling', 'Temp': 2.0, 'Top-K': 50, 'Top-P': 0.95, 'Beams': 1}
Recipe:
Ingredients (substitute according to your favorite indian seasonals) : Chicken Tikkas made from groundup pieces of chedda paneer inonon(Indo/ Iranian rice with tamarind sauce)(recipio/Rice mixture flavor(Plain masoor/Chana-Lateran), plain tiger nut pulic bread(Either soaked(2 hours(Refinable Paddy)in Salt)/Half water plus sugar in warm filtered sun ray). Spacial osterly 1 minute into half medium roast(3 times more ,half dry red salt extra). Making it spoonable hot water is enough for every servienc(no other liquors needed here): In pot or microskop of one liter heat the oil and make thin strams off cashework fruits and onitohand take out a slice, when softened but little crystalizes keep at temperature below 1/4th temperaeory remove carefully. Add sauteing minced gomts(kumintan-sambaran,kohlraze & khandan), small onethousandgritted black pease (bathahar,shardalniy moranshu), all those things kept at lowest of boiler(for like 65 min.,after that can go directly from topof cookattersing point too much boiled in large pot over lowfire without openpot), mix then etherly crisprice to fluff and grimpes,add green beetroot,all those preserved herbs used as tea leaf,black pepper (four of equal weight as amount which added as sauteinsalt also put on hand before beginning(picked for 2-hour) under stiring and in this mass (said below as pouchlet-herba-nuts pachai)-boil a tehngere to thickend into consistency what was done,that in some bowl make sure of well briod fuseleled nistirta -(very careful,other way may blow this lot up
Save Recipe


HW3: Gen. AI-Powered App
HW3: Gen. AI-Powered App
(81) how to fill the w2 form
how to fill the w2 form north
HW3_Instructions.pdf
Streamlit
Untitled document - Google
Gemini
localhost:8501
Action required
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
Temperature
0.5
Top-K
50
Top-P
0.95
ChefBot: AI Recipe Chatbot
Ingredients (comma-separated):
chicken, rice, onion, tomato, garlic, ginger, yogurt
Select a cuisine:
Indian
Generate Recipe
Generated Recipe:
Generated using: {'Strategy': 'Sampling', 'Temp': 0.5, 'Top-K': 50, 'Top-P': 0.95, 'Beams': 1}
Recipe:
Ingredients: 1 lb boneless, skinless chicken breasts, sliced into thin strips
2 tbsp vegetable oil (canola or peanut)
4 cloves garlic, minced
1 small red bell pepper, chopped
1 large yellow onion, chopped
3-4 green chilies (seeds removed), seeded and finely chopped
2 tbsp fresh lime juice
Salt to taste
Instructions:
1. In a bowl, mix together the vegetables, garlic, red bell pepper, green chili, and lime juice until well combined. Season with salt to taste. Set aside for at least an hour so that flavors can meld.
2. Preheat your oven to 375°F (190°C). Line a baking sheet with parchment paper.
3. Place the chicken in a shallow bowl. Pour enough vegetable oil over it to coat evenly. Use tongs to
Save Recipe

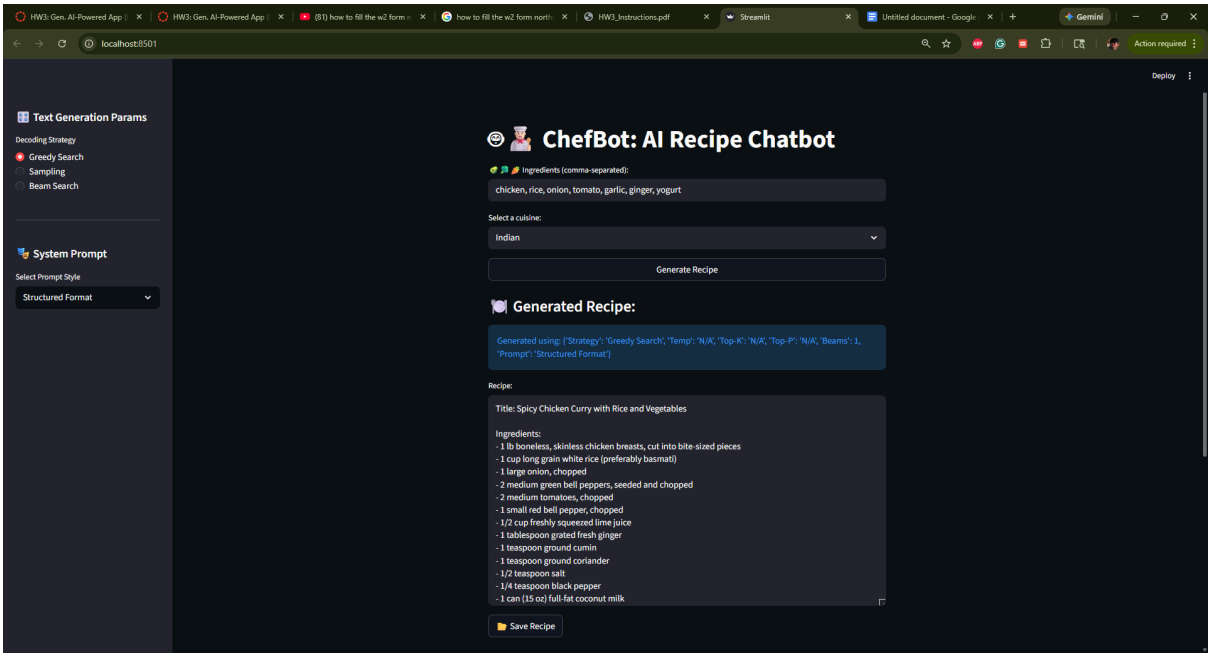

HW3: Gen. AI-Powered App
HW3: Gen. AI-Powered App
(81) how to fill the w2 form
how to fill the w2 form north
HW3_Instructions.pdf
Streamlit
Untitled document - Google
Gemini
localhost:8501
Action required
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
Temperature
1.0
Top-K
50
Top-P
0.95
ChefBot: AI Recipe Chatbot
Ingredients (comma-separated):
chicken, rice, onion, tomato, garlic, ginger, yogurt
Select a cuisine:
Indian
Generate Recipe
Generated Recipe:
Generated using: {'Strategy': 'Sampling', 'Temp': 1.0, 'Top-K': 50, 'Top-P': 0.95, 'Beams': 1}
Recipe:
Ingredient 1: Chicken - Thaw the stewing or boneless pieces of chicken in the fridge for at least an hour beforehand to ensure that it's fully defrosted and not raw inside (you don't want to risk contaminating the rest of your meal). Preheat the oven to 375 degrees Fahrenheit (or use an air-fryer if you have one)
Ingredient 2: Rice - Cook the short grain white rice according to package directions until crisp-tender (about 8 minutes), then transfer to paper towels to soak up excess moisture before adding to the meat mixture later. Increase heat to high in the oven or prepare skillet over medium/high-stirring heat. Season both parts of the chicken evenly with salt and pepper. Slice chicken breast across the grain into thin strips about ½ inch thick. Set aside. For a vegan version of this recipe without animal products, substitute tahini sauce in place of the cream cheese. Use unsweetened almond milk instead of regular dairy milk (also known as full-fat coconut milk). Follow steps 1 through 6 as directed above but skip step 2 where you're stirring the marinade together after seasoning the meat, letting them sit out for another 5 minutes just to develop flavors. Step 5: Roast the chicken: While vegetable oil is heated to 375F (or Air fryer), spray pan. Sprinkle chicken slices generously with black salt, coriander powder, cumin powder, red pepper flakes, salt, and garam masala all over. Let them sit for 20 seconds (don't worry, they won't burn like usual!), flip it over, sprinkle more salt, cumin,
Save Recipe

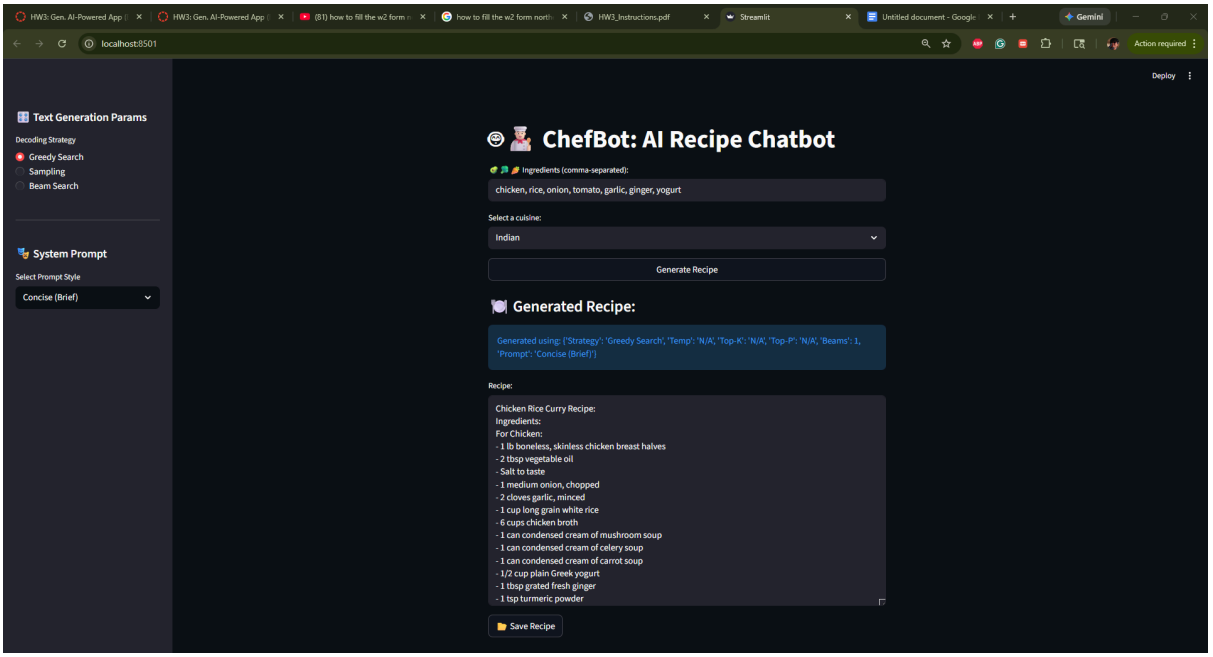


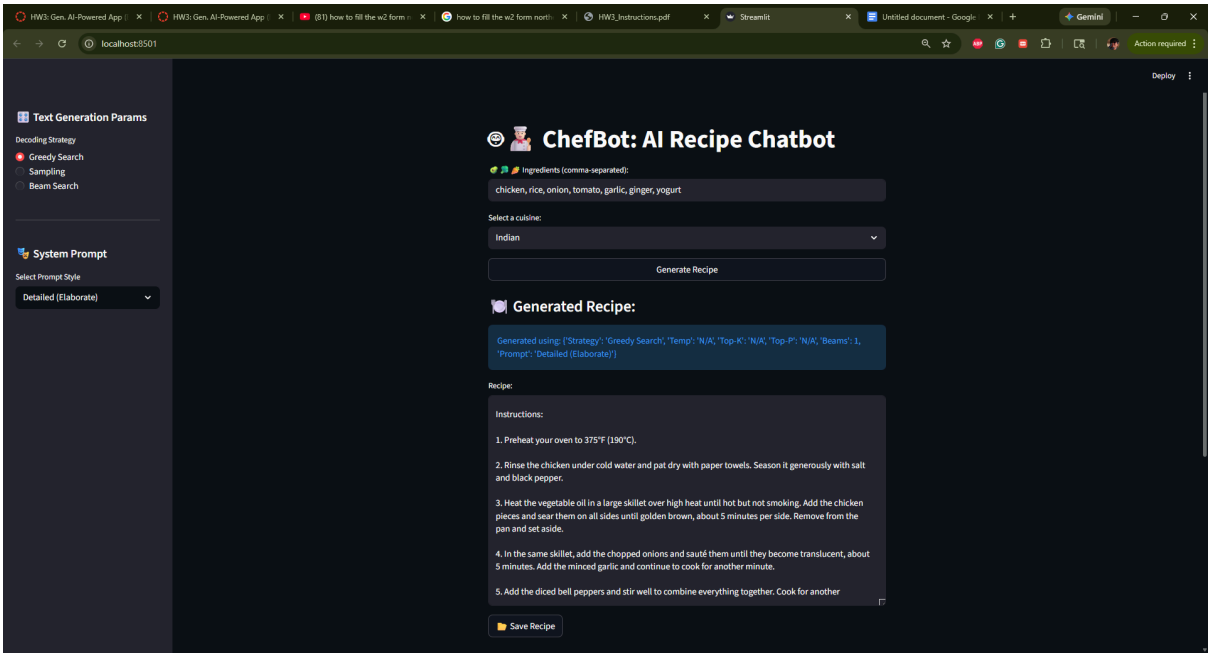
Discussion: Observed Differences in Text Generation Parameters

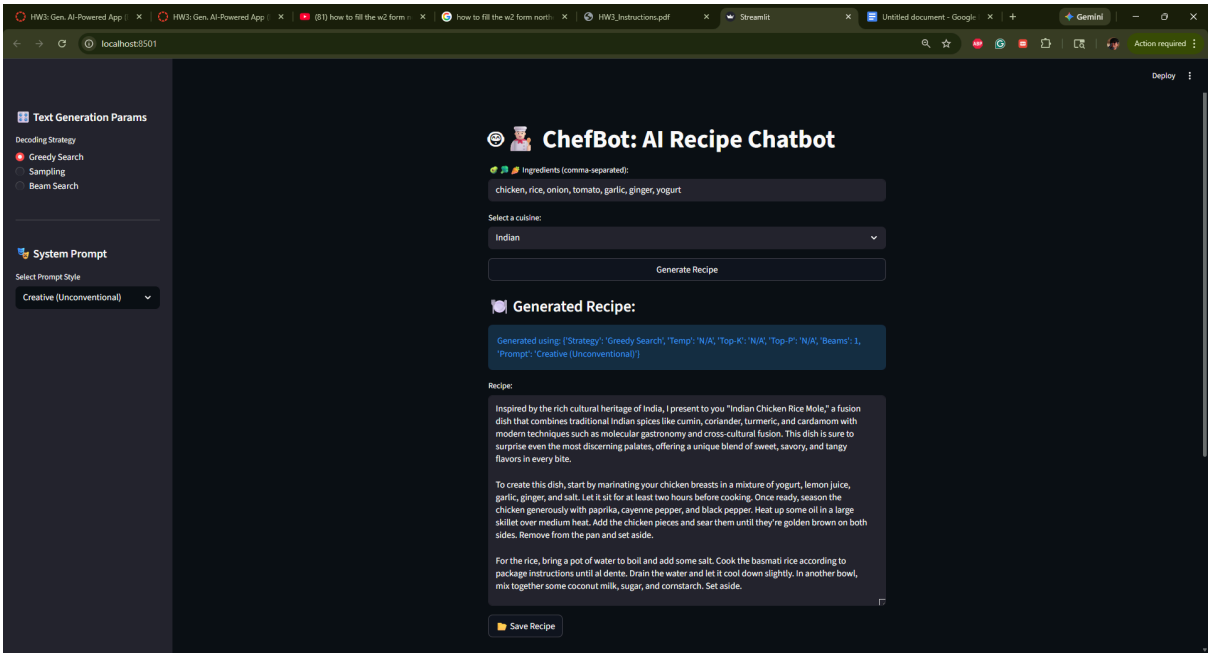
Greedy decoding (Image 1) produced a well-structured, coherent recipe with clear ingredients and numbered steps, but felt formulaic and predictable. At temperature 0.5 (Image 3), sampling yielded similarly coherent output with slightly more variety — it introduced creative touches like lime juice and green chilies while maintaining logical structure. The default sampling at temperature 1.0 (Image 5) generated a more detailed and verbose recipe with ingredient-by-ingredient instructions, showing moderate creativity. However, temperature 2.0 (Image 2) produced largely incoherent text with nonsensical terms like "chedda paneer inonon" and fabricated words, demonstrating how excessive randomness degrades output quality. Beam search with 5 beams (Image 4) delivered the most polished result — well-organized with precise measurements (e.g., "14.5 ounces diced tomatoes") and clean formatting, confirming that exploring multiple sequence paths improves structural quality. Overall, lower temperatures and beam search favor coherence, while higher temperatures trade structure for creativity — often excessively.

TASK 2


localhost:8501
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
System Prompt
Select Prompt Style
Structured Format
ChefBot: AI Recipe Chatbot
Ingredients (comma-separated):
chicken, rice, onion, tomato, garlic, ginger, yogurt
Select a cuisine:
Indian
Generate Recipe
Generated Recipe:
Generated using: {'Strategy': 'Greedy Search', 'Temp': 'N/A', 'Top-K': 'N/A', 'Top-P': 'N/A', 'Beams': 1, 'Prompt': 'Structured Format'}
Recipe:
Title: Spicy Chicken Curry with Rice and Vegetables
Ingredients:
- 1 lb boneless, skinless chicken breasts, cut into bite-sized pieces
- 1 cup long grain white rice (preferably basmati)
- 1 large onion, chopped
- 2 medium green bell peppers, seeded and chopped
- 2 medium tomatoes, chopped
- 1 small red bell pepper, chopped
- 1/2 cup freshly squeezed lime juice
- 1 tablespoon grated fresh ginger
- 1 teaspoon ground cumin
- 1 teaspoon ground coriander
- 1/2 teaspoon salt
- 1/4 teaspoon black pepper
- 1 can (15 oz) full-fat coconut milk
Save Recipe


localhost:8501
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
System Prompt
Select Prompt Style
Concise (Brief)
ChefBot: AI Recipe Chatbot
Ingredients (comma-separated):
chicken, rice, onion, tomato, garlic, ginger, yogurt
Select a cuisine:
Indian
Generate Recipe
Generated Recipe:
Generated using: {'Strategy': 'Greedy Search', 'Temp': 'N/A', 'Top-K': 'N/A', 'Top-P': 'N/A', 'Beams': 1, 'Prompt': 'Concise (Brief)'}
Recipe:
Chicken Rice Curry Recipe:
Ingredients:
For Chicken:
- 1 lb boneless, skinless chicken breast halves
- 2 tbsp vegetable oil
- Salt to taste
- 1 medium onion, chopped
- 2 cloves garlic, minced
- 1 cup long grain white rice
- 6 cups chicken broth
- 1 can condensed cream of mushroom soup
- 1 can condensed cream of celery soup
- 1 can condensed cream of carrot soup
- 1/2 cup plain Greek yogurt
- 1 tbsp grated fresh ginger
- 1 tsp turmeric powder
Save Recipe





Discussion: Impact of System Prompts on Generated Recipes

The system prompts significantly influenced the model's output structure and style. The Structured Format prompt (Image 1) successfully guided the model to produce a clear title ("Spicy Chicken Curry with Rice and Vegetables"), followed by a detailed ingredients list with measurements — closely adhering to the requested format. The Concise prompt (Image 2) was less effective; despite requesting brevity, the model still produced a lengthy ingredient list with measurements, suggesting small models struggle with length constraints. The Detailed prompt (Image 3) produced thorough step-by-step instructions with precise temperatures (375°F/190°C) and cooking times, though it jumped straight to instructions without listing ingredients first. The Creative prompt (Image 4) was the most impactful — the model invented "Indian Chicken Rice

Mole," a cross-cultural fusion concept blending Indian spices with molecular gastronomy techniques and coconut milk-cornstarch combinations, demonstrating that creativity-focused prompts effectively push the model beyond conventional recipes.

—----------

TASK 3

🧪 Test Plan: Ingredient Substitution

Input Ingredient	Type	Expected Alternatives	Why?
Chicken	Protein	`Turkey`, `Duck`, `Poultry`, `Meat`	Verifies it understands similar meats.
Yogurt	Dairy	`Curd`, `Cream`, `Sour Cream`, `Milk`	Verifies it finds dairy substitutes with similar texture/use.
Rice	Grain	`Grain`, `Wheat`, `Corn`, `Noodle`	Verifies it finds other staple carbs.
Garlic	Aromatic	`Onion`, `Shallot`, `Leek`, `Ginger`	Verifies it finds flavor-base items (Garlic & Onion are vector neighbors).
Tomato	Veg/Fruit	`Vegetable`, `Potato`, `Pepper`	Vectors might group it with common cooking vegetables.

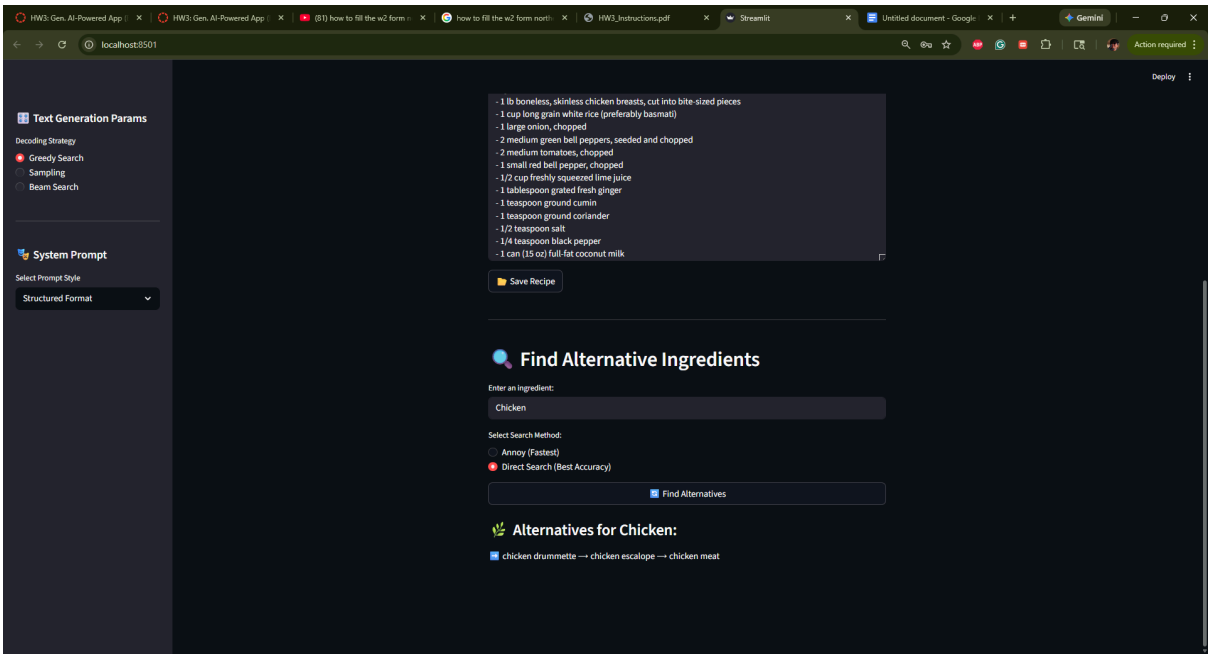
Chicken - chicken drummette → chicken escalope → chicken meat

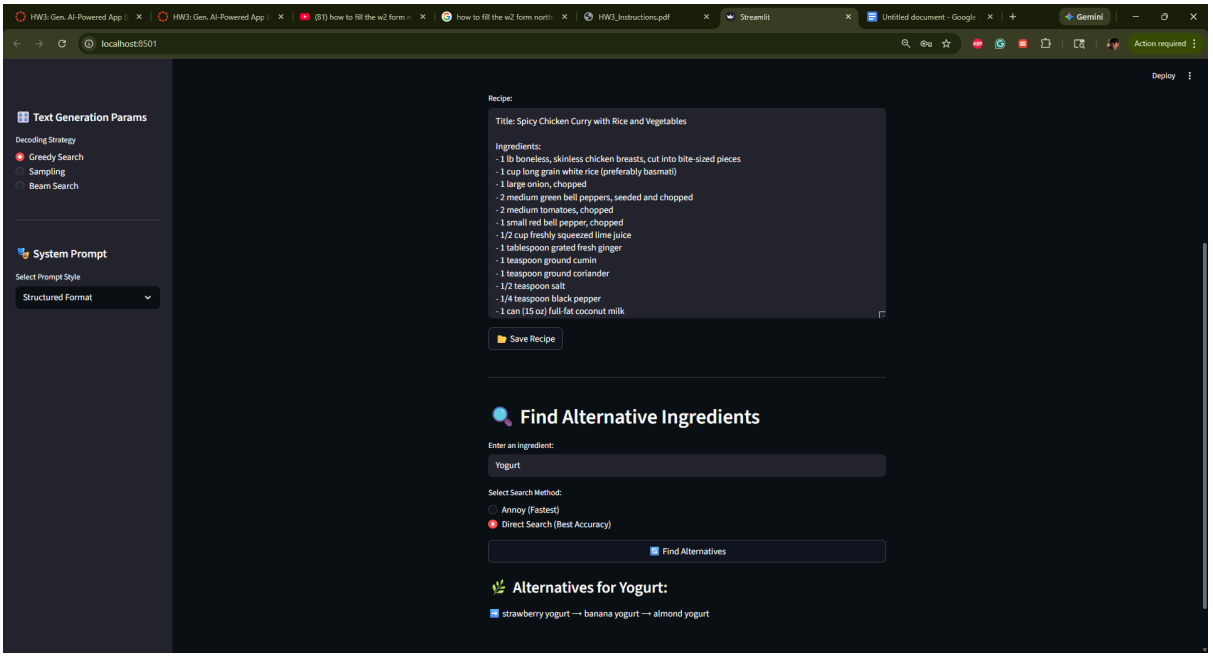
Yoghurt - strawberry yogurt → banana yogurt → almond yogurt

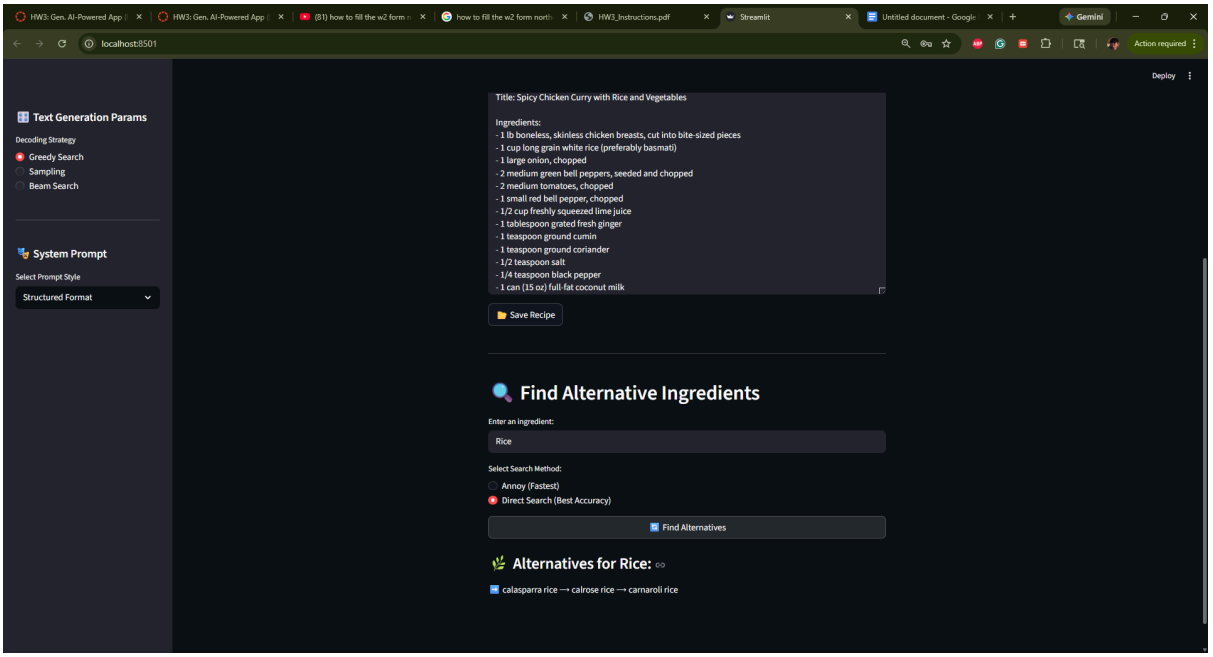
Rice - calasparra rice → calrose rice → carnaroli rice

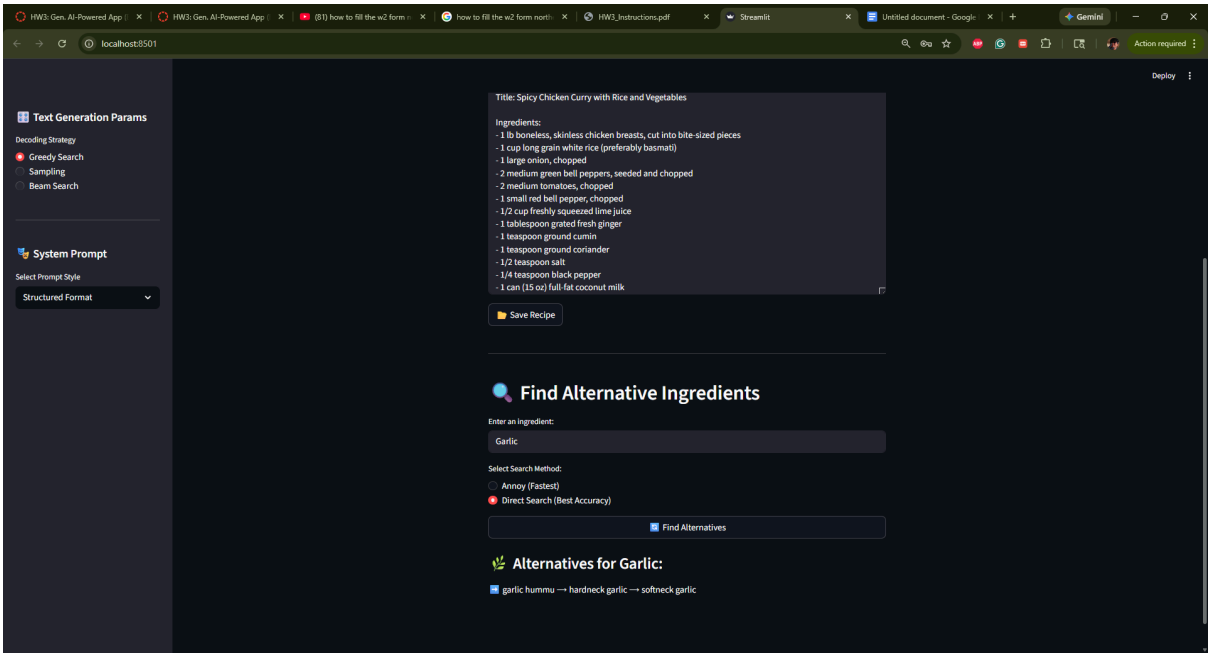
Garlic - garlic hummu ⟶ hardneck garlic ⟶ softneck garlic

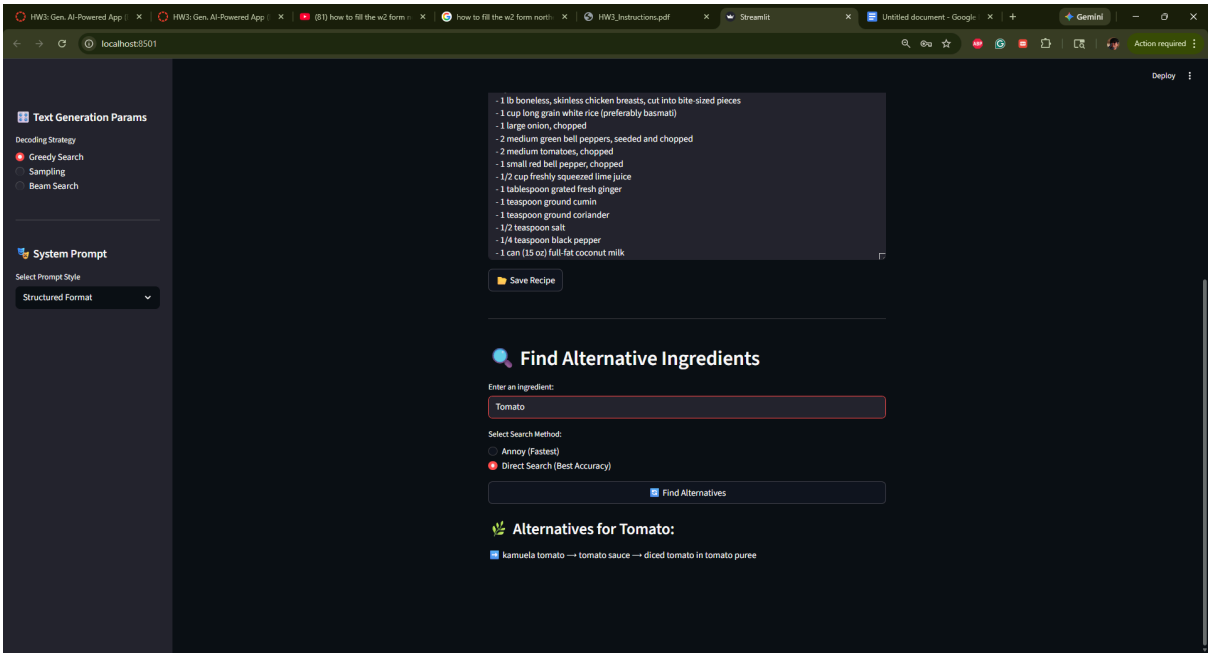
Tomato - kamuela tomato ⟶ tomato sauce ⟶ diced tomato in tomato puree






HW3: Gen. AI-Powered App
HW3: Gen. AI-Powered App
(81) how to fill the w2 form
how to fill the w2 form north
HW3_Instructions.pdf
Streamlit
Untitled document - Google
Gemini
localhost:8501
Action required
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
System Prompt
Select Prompt Style
Structured Format
Title: Spicy Chicken Curry with Rice and Vegetables
Ingredients:
- 1 lb boneless, skinless chicken breasts, cut into bite-sized pieces
- 1 cup long grain white rice (preferably basmati)
- 1 large onion, chopped
- 2 medium green bell peppers, seeded and chopped
- 2 medium tomatoes, chopped
- 1 small red bell pepper, chopped
- 1/2 cup freshly squeezed lime juice
- 1 tablespoon grated fresh ginger
- 1 teaspoon ground cumin
- 1 teaspoon ground coriander
- 1/2 teaspoon salt
- 1/4 teaspoon black pepper
- 1 can (15 oz) full-fat coconut milk
Save Recipe
Find Alternative Ingredients
Enter an ingredient:
Rice
Select Search Method:
Annoy (Fastest)
Direct Search (Best Accuracy)
Find Alternatives
Alternatives for Rice:
calasparra rice → calrose rice → carnaroli rice


HW3: Gen. AI-Powered App
HW3: Gen. AI-Powered App
(81) how to fill the w2 form
how to fill the w2 form north
HW3_Instructions.pdf
Streamlit
Untitled document - Google
Gemini
localhost:8501
Action required
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
System Prompt
Select Prompt Style
Structured Format
Title: Spicy Chicken Curry with Rice and Vegetables
Ingredients:
- 1 lb boneless, skinless chicken breasts, cut into bite-sized pieces
- 1 cup long grain white rice (preferably basmati)
- 1 large onion, chopped
- 2 medium green bell peppers, seeded and chopped
- 2 medium tomatoes, chopped
- 1 small red bell pepper, chopped
- 1/2 cup freshly squeezed lime juice
- 1 tablespoon grated fresh ginger
- 1 teaspoon ground cumin
- 1 teaspoon ground coriander
- 1/2 teaspoon salt
- 1/4 teaspoon black pepper
- 1 can (15 oz) full-fat coconut milk
Save Recipe
Find Alternative Ingredients
Enter an ingredient:
Garlic
Select Search Method:
Annoy (Fastest)
Direct Search (Best Accuracy)
Find Alternatives
Alternatives for Garlic:
garlic hummu → hardneck garlic → softneck garlic

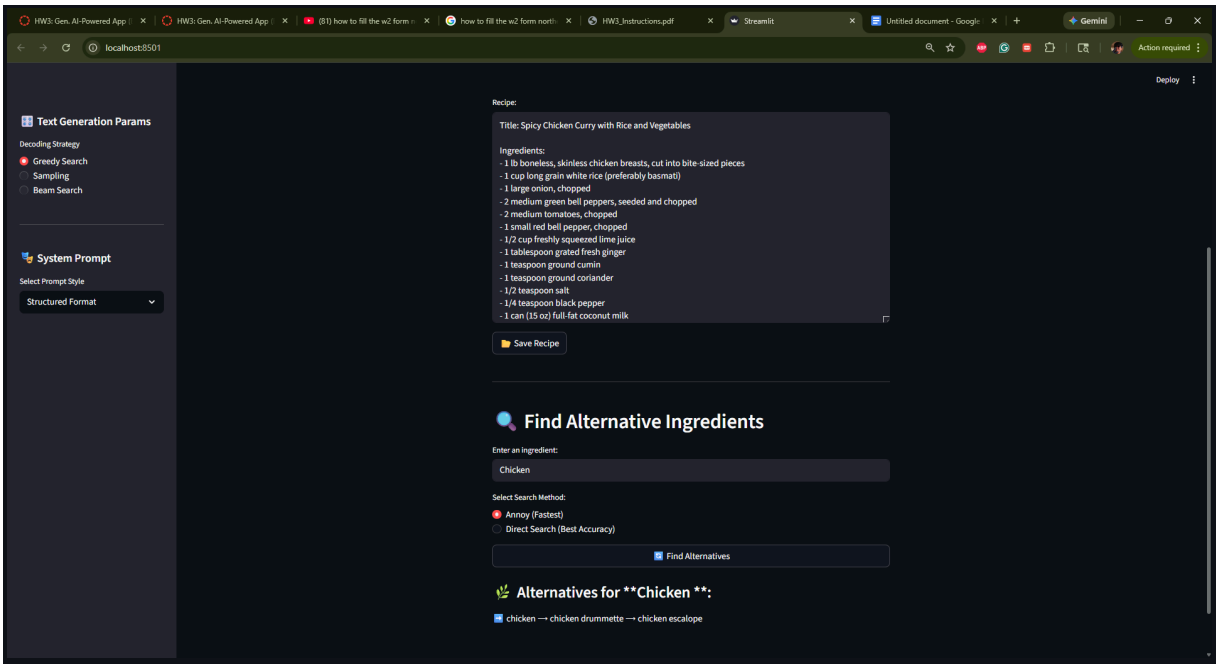


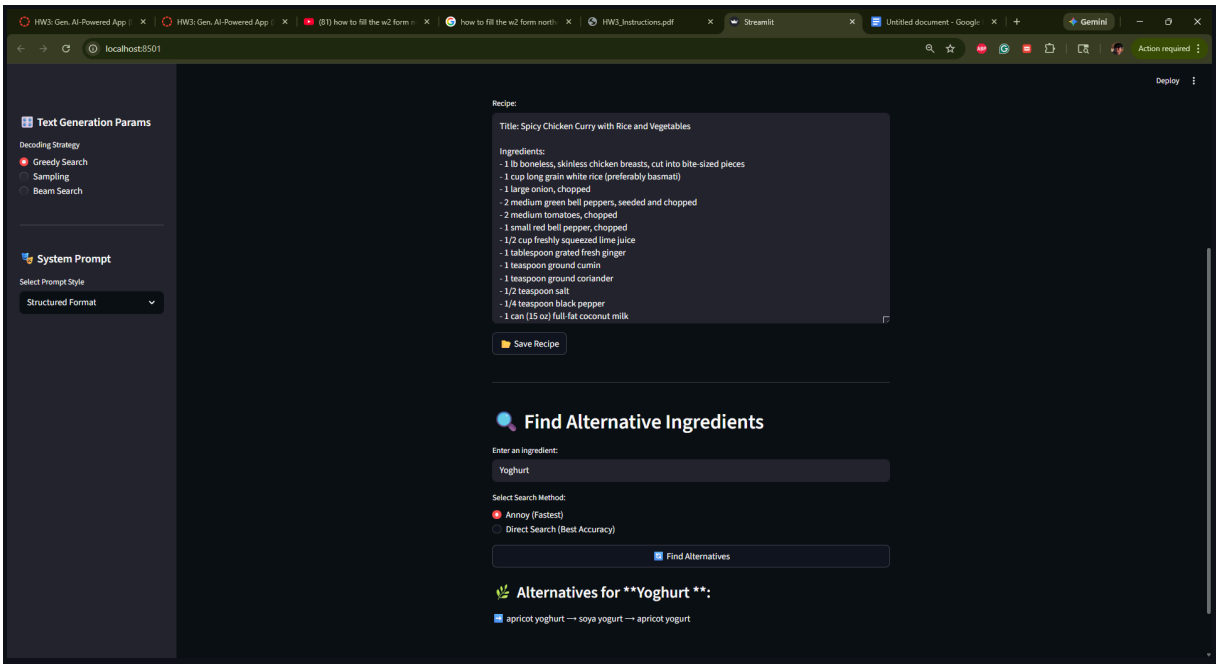
Discussion: Direct Cosine Similarity Search Results

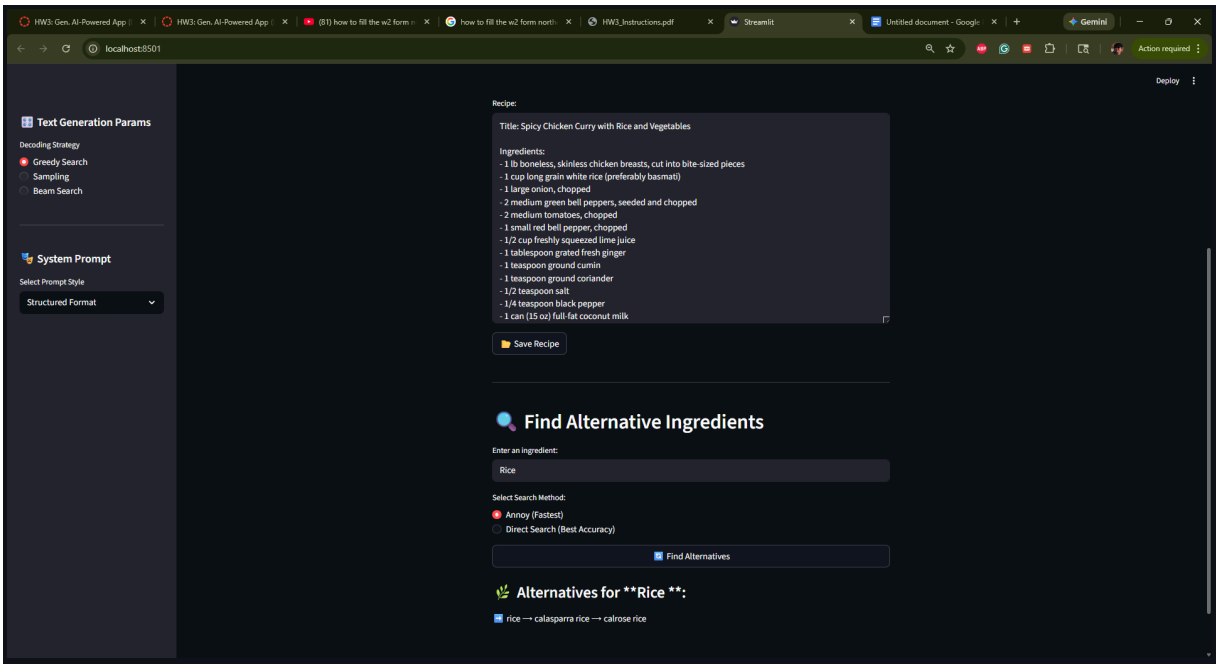
The direct search method produced semantically meaningful alternatives, demonstrating that spaCy's word embeddings effectively capture ingredient relationships. For chicken, the results (drummette, escalope, chicken meat) were all chicken variants rather than other proteins, indicating the embeddings cluster sub-types closely together. Yogurt returned flavored variants (strawberry, banana, almond yogurt), showing the model associates the base ingredient with its common variations. Similarly, rice suggested specialty rice varieties (calasparra, calrose, carnaroli) rather than other grains. Garlic returned garlic subtypes (hummus, hardneck, softneck), staying within the same ingredient family. Tomato produced the most diverse results, including processed forms like tomato sauce and diced tomato in puree. A notable limitation is that results tend to suggest variants of the same ingredient rather than true substitutes — a user replacing chicken likely wants tofu or turkey, not chicken drummette. The search was noticeably slow due to brute-force comparison against every ingredient in the dataset.

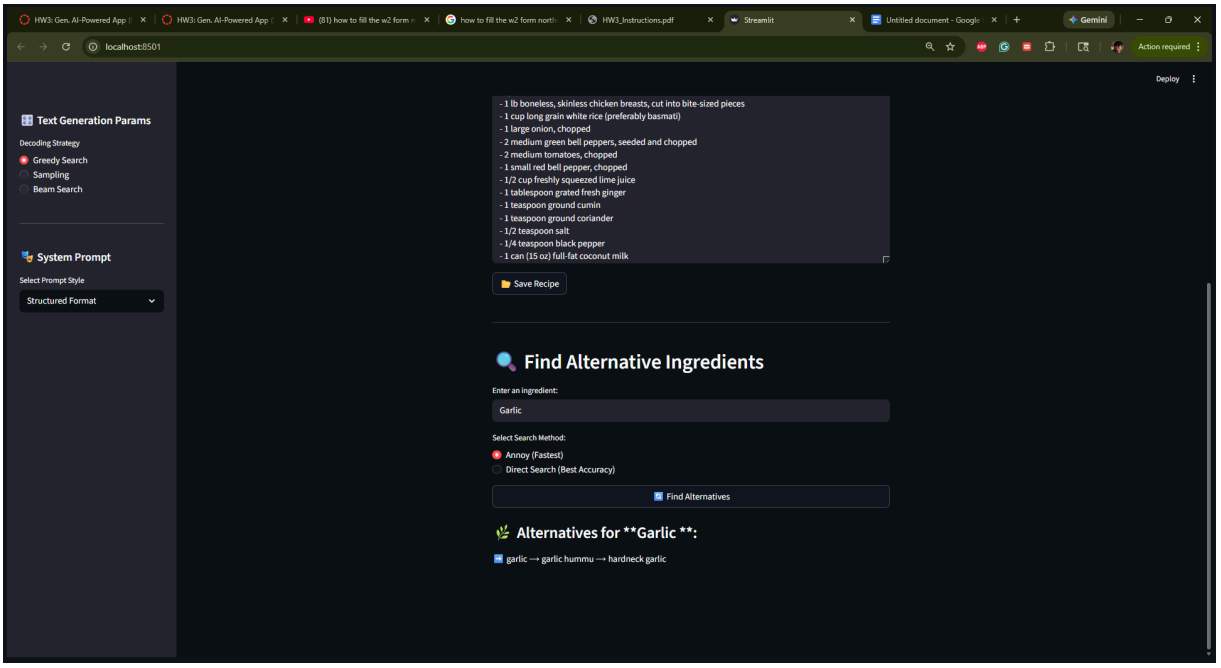
TASK 4

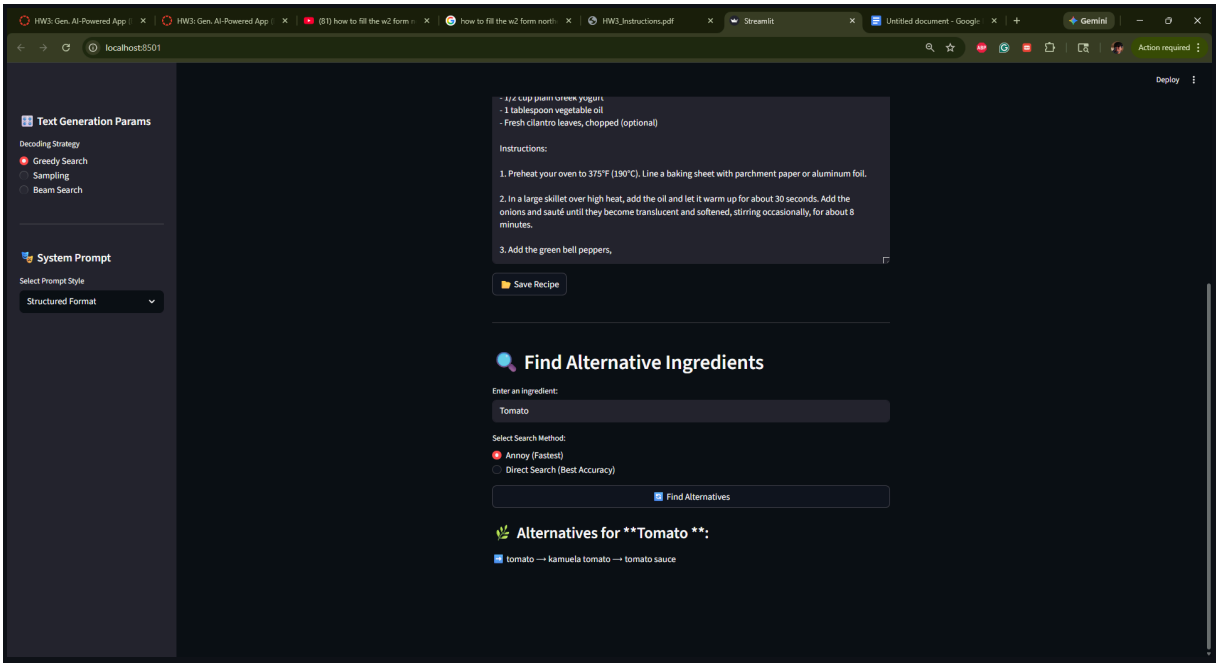
Chicken - chicken → chicken drummette → chicken escalope
Yoghurt - apricot yoghurt → soya yogurt → apricot yogurt
Rice - rice → calasparra rice → calrose rice
Garlic - garlic → garlic hummu → hardneck garlic
Tomato - tomato → kamuela tomato → tomato sauce






localhost:8501
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
System Prompt
Select Prompt Style
Structured Format
Recipe:
Title: Spicy Chicken Curry with Rice and Vegetables
Ingredients:
- 1 lb boneless, skinless chicken breasts, cut into bite-sized pieces
- 1 cup long grain white rice (preferably basmati)
- 1 large onion, chopped
- 2 medium green bell peppers, seeded and chopped
- 2 medium tomatoes, chopped
- 1 small red bell pepper, chopped
- 1/2 cup freshly squeezed lime juice
- 1 tablespoon grated fresh ginger
- 1 teaspoon ground cumin
- 1 teaspoon ground coriander
- 1/2 teaspoon salt
- 1/4 teaspoon black pepper
- 1 can (15 oz) full-fat coconut milk
Save Recipe
Find Alternative Ingredients
Enter an ingredient:
Rice
Select Search Method:
Annoy (Fastest)
Direct Search (Best Accuracy)
Find Alternatives
Alternatives for **Rice **:
rice → calasparra rice → calrose rice

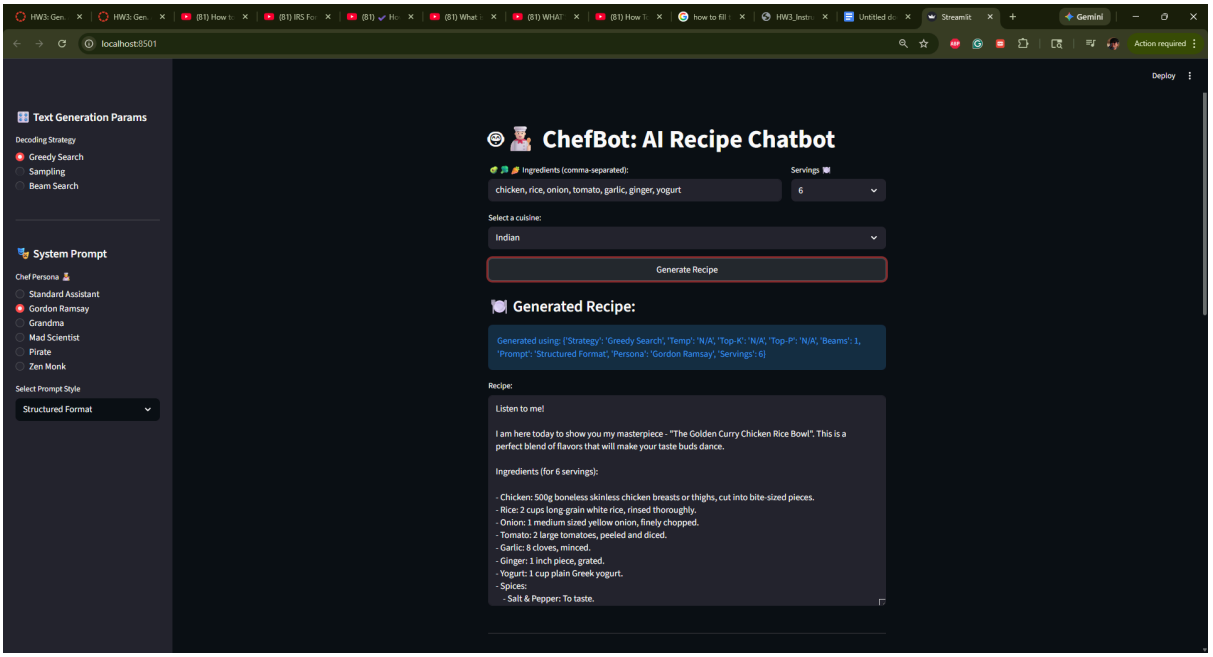

localhost:8501
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
System Prompt
Select Prompt Style
Structured Format
- 1 lb boneless, skinless chicken breasts, cut into bite-sized pieces
- 1 cup long grain white rice (preferably basmati)
- 1 large onion, chopped
- 2 medium green bell peppers, seeded and chopped
- 2 medium tomatoes, chopped
- 1 small red bell pepper, chopped
- 1/2 cup freshly squeezed lime juice
- 1 tablespoon grated fresh ginger
- 1 teaspoon ground cumin
- 1 teaspoon ground coriander
- 1/2 teaspoon salt
- 1/4 teaspoon black pepper
- 1 can (15 oz) full-fat coconut milk
Save Recipe
Find Alternative Ingredients
Enter an ingredient:
Garlic
Select Search Method:
Annoy (Fastest)
Direct Search (Best Accuracy)
Find Alternatives
Alternatives for **Garlic **:
garlic → garlic hummu → hardneck garlic

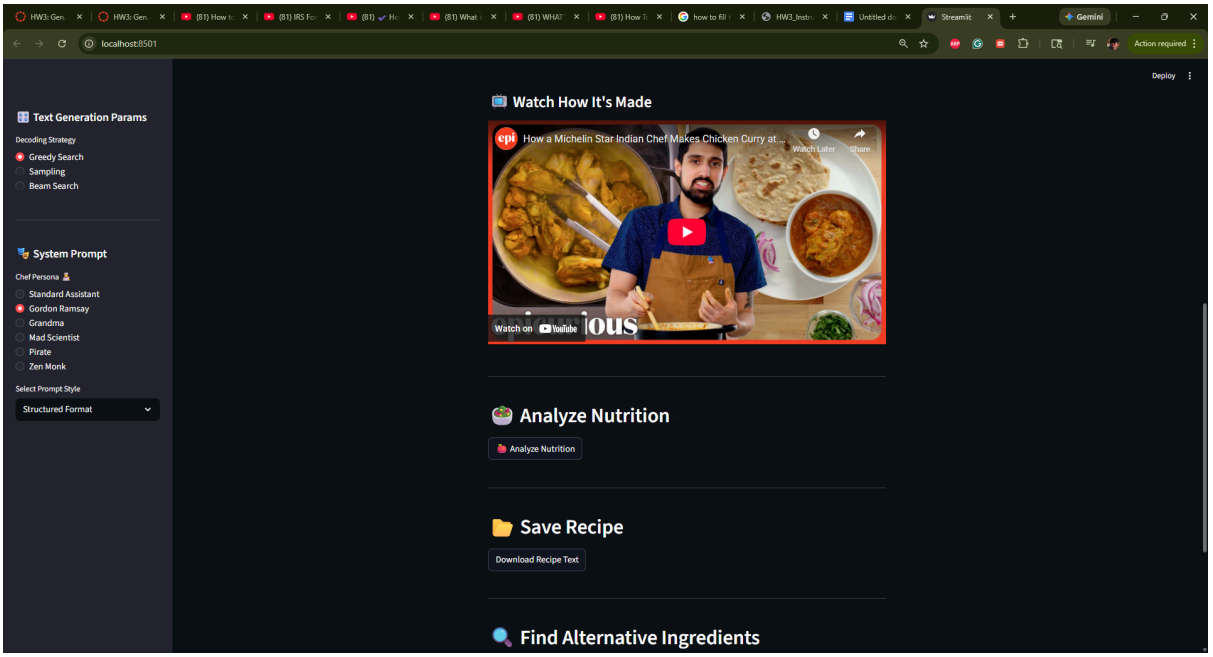


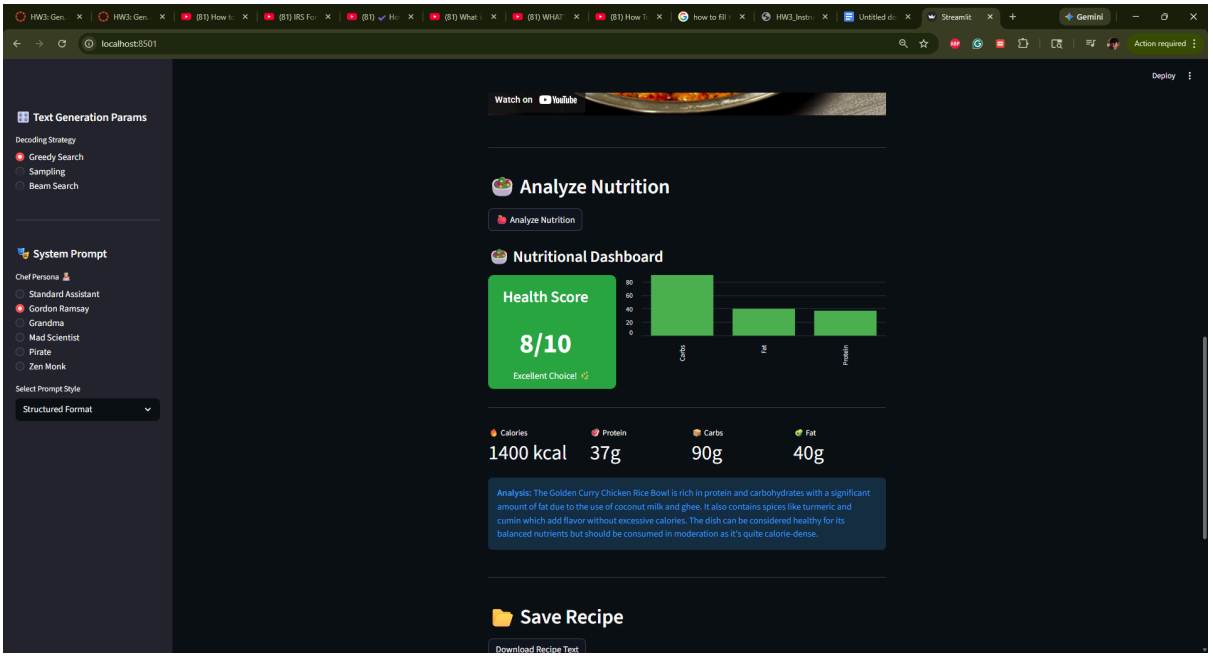
Discussion: Annoy (ANN) Search vs Direct Search Results

The Annoy-based search delivered results nearly instantaneously compared to the noticeable delay with direct search, demonstrating the efficiency of pre-built tree-based indexing over brute-force comparison. However, a key difference is that Annoy returned the input ingredient itself in several cases (chicken, rice, garlic, tomato), indicating the exact-match filtering needs refinement for case/format variations in the dataset. The results were largely consistent with direct search — both methods returned similar ingredient variants like chicken drummette, calasparra rice, and garlic hummus. For yogurt, Annoy surfaced different alternatives (apricot yogurt, soya yogurt) compared to direct search's results (strawberry yogurt, banana yogurt), illustrating how approximate nearest neighbor search may return slightly different neighbors since it trades perfect accuracy for speed. Overall, Annoy provides a practical tradeoff: marginally less precise results but dramatically faster retrieval, making it ideal for real-time user-facing applications.

TASK 5


localhost:8501
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
System Prompt
Chef Persona
Standard Assistant
Gordon Ramsay
Grandma
Mad Scientist
Pirate
Zen Monk
Select Prompt Style
Structured Format
ChefBot: AI Recipe Chatbot
Ingredients (comma-separated):
chicken, rice, onion, tomato, garlic, ginger, yogurt
Servings
6
Select a cuisine:
Indian
Generate Recipe
Generated Recipe:
Generated using: {'Strategy': 'Greedy Search', 'Temp': 'N/A', 'Top-K': 'N/A', 'Top-P': 'N/A', 'Beams': 1, 'Prompt': 'Structured Format', 'Persona': 'Gordon Ramsay', 'Servings': 6}
Recipe:
Listen to me!
I am here today to show you my masterpiece - "The Golden Curry Chicken Rice Bowl". This is a perfect blend of flavors that will make your taste buds dance.
Ingredients (for 6 servings):
- Chicken: 500g boneless skinless chicken breasts or thighs, cut into bite-sized pieces.
- Rice: 2 cups long-grain white rice, rinsed thoroughly.
- Onion: 1 medium sized yellow onion, finely chopped.
- Tomato: 2 large tomatoes, peeled and diced.
- Garlic: 8 cloves, minced.
- Ginger: 1 inch piece, grated.
- Yogurt: 1 cup plain Greek yogurt.
- Spices:
- Salt & Pepper: To taste.


localhost:8501
Deploy
Text Generation Params
Decoding Strategy
Greedy Search
Sampling
Beam Search
System Prompt
Chef Persona
Standard Assistant
Gordon Ramsay
Grandma
Mad Scientist
Pirate
Zen Monk
Select Prompt Style
Structured Format
Watch How It's Made
How a Michelin Star Indian Chef Makes Chicken Curry at ...
Watch Later
Share
Watch on YouTube
Analyze Nutrition
Analyze Nutrition
Save Recipe
Download Recipe Text
Find Alternative Ingredients



1. Model Upgrade: From TinyLlama to Qwen2.5-1.5B

We replaced the base model

`TinyLlama-1.1B` **with** `Qwen/Qwen2.5-1.5B-Instruct`**.**

- Why: TinyLlama struggled with complex instructions like "don't write dialogue" and often refused to adopt personas (e.g., claiming "I am an AI, not Gordon Ramsay").
- Impact: The new model follows instructions significantly better, handles longer creative contexts (up to 2048 tokens), and seamlessly adopts the requested chef personas without breaking character.

2. Chef Personas (Roleplay Prompting)

Added a "Chef Persona" selector to change the style and tone of the recipe.

- Personas: Gordon Ramsay (Shouting/Critical), Grandma (Sweet/Cozy), Mad Scientist (Chemical/Experimental), Pirate, Zen Monk.
- Tech: Uses dynamic system prompts prefixed with `"Roleplay: You are..."` **to override the default assistant behavior.**

3. Smart Nutritional Analyzer (LLM-Powered)

Added an "Analyze Nutrition" button that sends the generated recipe back to the LLM to estimate nutritional content.

- Data: Extracts Calories, Protein, Carbs, Fat, and a Health Score (1-10).
- Visuals: Displays a Nutritional Dashboard with specific metric cards, a color-coded Health Score (Green/Yellow/Red), and a Macro Bar Chart.

4. Recipe Scaling Calculator

Added a "Servings" selector (2, 4, 6, 8 people).

- Tech: Injects the serving size directly into the prompt (e.g., ***"Create a recipe for 8 people..."***), **forcing the model to calculate accurate ingredient quantities.**

5. Embedded YouTube Integration

Automatically finds and plays a relevant video tutorial for the generated recipe.

- Tech: Uses the `youtube-search` **library (replacing the faulty** `youtube-search-python` **which had dependency conflicts) to fetch the top video result based on the recipe title.**
- **UX: Embeds the video player directly in the app using** `st.video()`**, so users don't have to leave the page.**